



EXAM PAPERS PRACTICE

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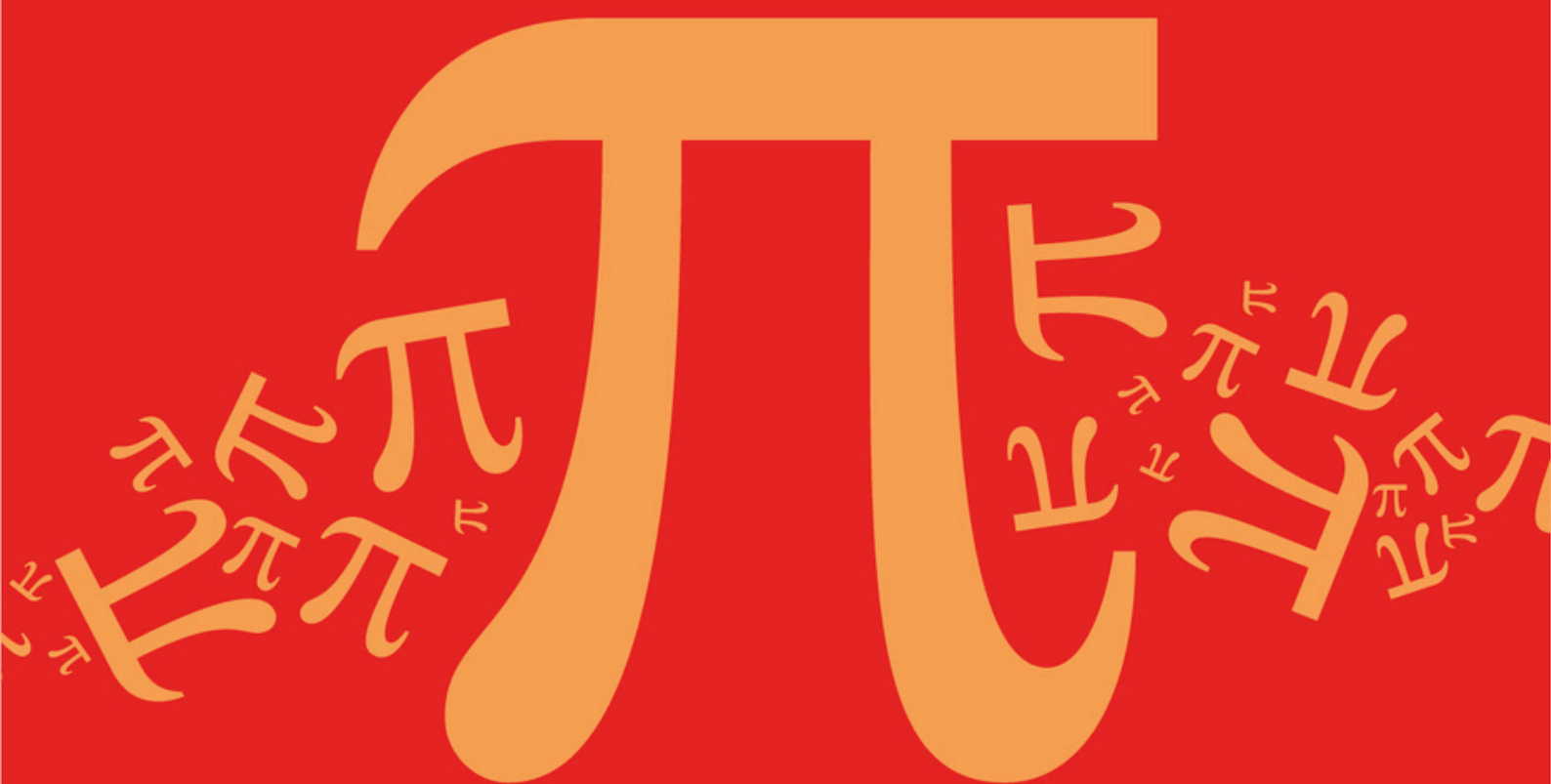
Practice questions created by actual examiners and assessment experts

Detailed mark scheme

Suitable for all boards

Designed to test your ability and thoroughly prepare you

AQA A Level Further Mathematics (7367)



Topic

B: Complex Numbers

Sub topic

B8: De Moivre's Theorem

Mark Scheme

AQA A Level Further Mathematics (7367)

Mark Scheme

Topic

B: Complex Numbers

Sub topic

B8: De Moivre's Theorem

EXAM PAPERS PRACTICE

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**To be used by all students and teachers for
AQA A Level Further Mathematics (7367)
for exams from 2025 onwards.**

Students and teachers of other boards may also find this useful

Mark schemes

Q1.

Marking Instructions	AO	Marks	Typical Solution
Uses De Moivre's theorem	AO3.1a	M1	$(\cos\theta + i\sin\theta)^5 = \frac{1}{\sqrt{2}}(1-i)$
Equates real and imaginary parts and obtains two equations	AO1.1a	A1	$\Rightarrow \cos 5\theta + i\sin 5\theta = \frac{1}{\sqrt{2}}(1-i)$
Deduces that the smallest possible value of 5θ is $\frac{7\pi}{4}$ FT from 'their' equations provided M1 has been awarded	AO2.2a	A1F	$\cos 5\theta = \frac{1}{\sqrt{2}}$ $\sin 5\theta = -\frac{1}{\sqrt{2}}$
Obtains the smallest possible value of θ from fully correct reasoning FT from 'their' 5θ provided M1 has been awarded	AO1.1b	A1F	$(5\theta =) \frac{7\pi}{4}$ $\theta = \frac{7\pi}{20}$
Total 4 marks			

Q2.

(a) (i) $\cos 4\theta + i \sin 4\theta = (\cos \theta + i \sin \theta)^4$

De Moivre & attempt to expand RHS

M1

$$\cos^4 \theta + 4i \cos^3 \theta \sin \theta + 6i^2 \cos^2 \theta \sin^2 \theta + 4i^3 \cos \theta \sin^3 \theta + i^4 \sin^4 \theta$$

any correct expansion

A1

Equating "their" real parts
or imaginary parts

m1

$$\cos 4\theta = \cos^4 \theta - 6\cos^2 \theta \sin^2 \theta + \sin^4 \theta$$

AG *be convinced*

A1

$$\sin 4\theta = 4\cos^3 \theta \sin \theta - 4\cos \theta \sin^3 \theta$$

correct

B1

5

$$(ii) \quad \tan 4\theta = \frac{\text{"their expression for " } \sin 4\theta}{\text{"their expression for " } \cos 4\theta}$$

M1

Division by $\cos^4\theta$

m1

$$\tan 4\theta = \frac{4\tan\theta - 4\tan^3\theta}{1 - 6\tan^2\theta + \tan^4\theta}$$

AG be convinced

A1

3

$$(b) \quad (\tan 4\theta = 1 \Rightarrow) \quad 1 = \frac{4t - 4t^3}{1 - 6t^2 + t^4}$$

when $\theta = \frac{\pi}{16}$

M1

$$1 - 6t^2 + t^4 = 4t - 4t^3 \Rightarrow t^4 + 4t^3 - 6t^2 - 4t + 1 = 0$$

AG be convinced

A1

$\frac{\pi}{16}$ 2025 Exam Papers Practice. All rights reserved.
 $\theta = \frac{\pi}{16}$ satisfies $\tan 4\theta = 1$

$\Rightarrow \tan \frac{\pi}{16}$ is root of quartic equation
both statements required

E1

(other roots are) $\tan \frac{5\pi}{16}, \tan \frac{9\pi}{16}, \tan \frac{13\pi}{16}$
or equivalent tan expressions

B1

4

$$(c) \quad \Sigma\alpha = -4 \quad \text{and} \quad \Sigma\alpha\beta = -6$$

watch for minus signs

B1

$$\Sigma\alpha^2 = (\Sigma\alpha)^2 - 2\Sigma\alpha\beta$$

correct formula

M1

$$(= 16 + 12) = 28$$

A1cso

$$\tan \frac{9\pi}{16} = -\tan \frac{7\pi}{16}, \quad \tan \frac{13\pi}{16} = -\tan \frac{3\pi}{16}$$

explicitly seen

B1

$$\tan^2 \frac{\pi}{16} + \tan^2 \frac{3\pi}{16} + \tan^2 \frac{5\pi}{16} + \tan^2 \frac{7\pi}{16} = 28$$

AG must earn previous 4 marks

A1cso

5

[17]

Q3.

$$\text{Numerator} = e^{\frac{pi}{8}}$$

$$\text{Denominator} = e^{\frac{-q\pi i}{12}}$$

$$\text{Fraction} = e^{\frac{pi}{8} + \frac{q\pi i}{12}}$$

allow for attempt to subtract powers

B1

B1

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M1

$$= e^{\frac{\pi i}{24} (3p + 2q)}$$

A1

$$i = e^{\frac{12\pi i}{24}}$$

OE

m1

$$3p + 2q = 12$$

ft errors of sign or arithmetic slips

A1F

$$p = 2, q = 3$$

CAO

A1

Alternative 1

$$\text{Numerator} = \cos \frac{p\pi}{8} + i \sin \frac{p\pi}{8}$$

(B1)

$$\text{Denominator} = \cos \frac{-q\pi}{12} + i \sin \frac{-q\pi}{12}$$

needs more than just $\cos \frac{q\pi}{12} - \sin \frac{p\pi}{8}$

(B1)

$$\text{Fraction} = \left(\cos \frac{p\pi}{8} + i \sin \frac{p\pi}{8} \right) \left(\cos \frac{q\pi}{12} + i \sin \frac{q\pi}{12} \right)$$

(M1)

$$= \cos \frac{\pi}{24}(3p + 2q) + i \sin \frac{\pi}{24}(3p + 2q)$$

(A1)

$$= i \text{ if } \cos \frac{\pi}{24}(3p + 2q) = 0 \text{ or } \sin \frac{\pi}{24}(3p + 2q) = 1$$

(m1)

$$3p + 2q = 12$$

(A1F)

$$p = 2, q = 3$$

CAO

(A1)

Alternative 2

$$\text{LHS} \cos \frac{p\pi}{8} + i \sin \frac{p\pi}{8}$$

(B1)

$$\text{RHS} i \cos \frac{q\pi}{12} + \sin \frac{q\pi}{12}$$

(B1)

$$\cos \frac{p\pi}{8} = \sin \frac{q\pi}{12} \text{ or } \sin \frac{p\pi}{8} = \cos \frac{q\pi}{12}$$

(M1)

$$\text{Introduction of } \frac{\pi}{2}$$

(m1)

$$\frac{p\pi}{8} = \frac{\pi}{2} - \frac{q\pi}{12}$$

(A1)

$$3p + 2q = 12$$

(A1F)

$$p = 2, q = 3$$

CAO (correct answers, insufficient working 3 / 7 only)

(A1)

[7]

Q4.

(a) $1, e^{\frac{2\pi i}{5}}, e^{\frac{4\pi i}{5}}, e^{\frac{-2\pi i}{5}}, e^{\frac{-4\pi i}{5}}$
accept e^{θ}

B1

1

(b) $\frac{z^5 - 1}{z - 1} = z^4 + z^3 + z^2 + z + 1$
B0 if assumed

B1

$$= \left(z - e^{\frac{2\pi i}{5}} \right) \left(z - e^{\frac{4\pi i}{5}} \right) \left(z - e^{\frac{-2\pi i}{5}} \right) \left(z - e^{\frac{-4\pi i}{5}} \right)$$

accept if $e^{\frac{6\pi i}{5}}, e^{\frac{8\pi i}{5}}$ used here

M1A1

3

(c) Correct grouping of linear factors

M1

$$e^{\frac{2\pi i}{5}} + e^{\frac{-2\pi i}{5}} = 2\cos \frac{2\pi}{5}$$

clearly shown

A1

$$\left(z^2 - 2\cos \frac{2\pi}{5} z + 1 \right) \left(z^2 - 2\cos \frac{4\pi}{5} z + 1 \right)$$

A1

$\div z^2$ to give answer

AG

A1

4

(d) Substitute into LHS to give $w^2 + w - 1$

B1

$$\text{RHS} \left(w - 2\cos \frac{2\pi}{5} \right) \left(w - 2\cos \frac{4\pi}{5} \right)$$

B1

Solve $w^2 + w - 1 = 0$

M1

$$w = \frac{-1 \pm \sqrt{5}}{2}$$

A1

$$\cos \frac{2\pi}{5} = \frac{\sqrt{5}-1}{4}$$

A1

with reasons for choice

E1

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6

[14]

Q5.

(a) Use of $(\cos \theta + i \sin \theta)^n = \cos n\theta + i \sin n\theta$

Stated or used

M1

$$\cos(-n\theta) + i \sin(-n\theta) = \cos n\theta - i \sin n\theta$$

*allow $\frac{2}{3}$ if this line is assumed
allow if complex conjugate used*

A1

$$z^n + \frac{1}{z^n} = 2\cos n\theta$$

AG

A1

3

- (b) (i) $z^8 + 4z^4 + 6 + 4z^{-4} + z^{-8}$
allow in retrospect

B1

1

- (ii) $z^2 + \frac{1}{z^2} = 2\cos 2\theta$ used
Can be implied from (b)(i)

B1

$$(2\cos 2\theta)^4 = 2\cos 8\theta + 8\cos 4\theta + 6$$

M1 for RHS

A1 for whole line

M1A1

$$\cos^4 2\theta = \frac{1}{8} \cos 8\theta + \frac{1}{2} \cos 4\theta + \frac{3}{8}$$

ft coefficients on previous line

A1F

4

Alternative:

$$\cos^4 2\theta = \left(\frac{1 + \cos 4\theta}{2} \right)^2$$

(M1)

(A1)

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$$\cos^2 4\theta = \frac{1}{2}(1 + \cos 8\theta)$$

(B1)

Final result

(A1)

- (c) $8\cos^4 2\theta = \cos 8\theta + 5 \rightarrow \cos 4\theta = \frac{1}{2}$
ft provided simplifies to $\cos 4\theta = p$

M1

A1F

$$k = \frac{1}{12}, \frac{5}{12}, \frac{7}{12}, \frac{11}{12}$$

CAO

A1

3

(d) $\int_0^{\frac{\pi}{2}} \cos^4 2\theta \, d\theta = \left[\frac{\sin 8\theta}{64} + \frac{\sin 4\theta}{8} + \frac{3}{8}\theta \right]_0^{\frac{\pi}{2}}$
ie their $\cos^4 2\theta$

M1
A1F

$$= \frac{3\pi}{16}$$

AG

A1

3

[14]

Q6.

(a) (i) $\cos 5\theta + i \sin 5\theta = (\cos \theta + i \sin \theta)^5$
Attempt to expand 3 correct terms

M1

Expansion in any form
 Correct simplification

A1

Equate real parts:

m1

$$\cos 5\theta = \cos^5 \theta - 10 \cos^3 \theta \sin^2 \theta + 5 \cos \theta \sin^4 \theta$$

AG

A1

Equate imaginary parts:

$$\sin 5\theta = 5 \cos^4 \theta \sin \theta - 10 \cos^2 \theta \sin^3 \theta + \sin^5 \theta$$

CAO

A1

5

(ii) $\tan 5\theta = \frac{\sin 5\theta}{\cos 5\theta}$
Used

M1

Division by $\cos^5 \theta$ or by $\cos^4 \theta$

m1

$$\tan 5\theta = \frac{\tan \theta (5 - 10 \tan^2 \theta + \tan^4 \theta)}{1 - 10 \tan^2 \theta + 5 \tan^4 \theta}$$

AG

A1

3

(b) $\theta = \frac{\pi}{5} \Rightarrow \tan 5\theta = 0$

Or for $\tan^4 \theta - 10 \tan^2 \theta + 5 = 0$

M1

$\therefore \tan \frac{\pi}{5}$ satisfies $t^4 - 10t^2 + 5 = 0$

Or for $\tan 5\theta = 0$

A1

Other roots $\tan \frac{k\pi}{5} \quad k = 2, 3, 4$

OE

B1

3

(c) Product of roots = 5

M1

$\tan \frac{\pi}{5} = -\tan \frac{4\pi}{5}$

Or $\tan \frac{2\pi}{5} = -\tan \frac{3\pi}{5}$

B1

$\tan^2 \frac{\pi}{5} \tan^2 \frac{2\pi}{5} = 5$

A1

$\tan \frac{\pi}{5} \tan \frac{2\pi}{5} = +\sqrt{5}$

A1

– sign rejected with reason

E1

5

Alternative

Use of quadratic formula M1

$$t^2 = 5 \pm 2\sqrt{5} \quad A1$$

$$t = \pm\sqrt{5 \pm 2\sqrt{5}} \quad B1$$

Correct selection of +ve values E1

Multiplied together to get $\sqrt{5}$ A1

[16]

Q7.

(a) (i) $\beta = 2 - 2\sqrt{3}i$

B1

1

(ii) $\alpha\beta\gamma = -8$
 Allow for +8 but not ± 16

M1

$$\alpha\beta = 16$$

B1

$$\gamma = -\frac{1}{2}$$

A1

3

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(iii) Either $\frac{-p}{2} = \alpha + \beta + \gamma$

or $\frac{q}{2} = \alpha\beta + \beta\gamma + \gamma\alpha$

SC if failure to divide by 2 throughout,
 allow M1A1 for either p or q correct ft

M1

$$p = -7, \quad q = 28$$

ft incorrect γ

A1F,A1F

3

Alternative to (a)(ii) and (a)(iii)

$$(z^2 - 4z + 16)(az + b)$$

(M1)

$$\alpha\beta = 16$$

(B1)

$$a = 2, b = +1, \gamma = -\frac{1}{2}$$

(A1)

Equating coefficients

(M1)

$$p = -7$$

(A1F)

$$q = 28$$

(A1F)

(b) (i) $r = 4, \theta = \frac{\pi}{3}$

B1,B1

2

(ii) $(2 + 2\sqrt{3}i)^n = \left(4e^{i\frac{\pi}{3}}\right)^n$

M1

$$= 4^n \left(\cos \frac{2\pi}{3} + i \sin \frac{2\pi}{3} \right)$$

AG

A1

2

(iii) $(2 - 2\sqrt{3}i)^n = 4^n \left(\cos \frac{2\pi}{3} - i \sin \frac{2\pi}{3} \right)$

B1

$$\begin{aligned} \alpha^n + \beta^n + \gamma^n &= 4^n \left(\cos \frac{2\pi}{3} + i \sin \frac{2\pi}{3} \right) \\ &+ 4^n \left(\cos \frac{2\pi}{3} - i \sin \frac{2\pi}{3} \right) + \left(-\frac{1}{2} \right)^n \end{aligned}$$

M1

$$= 2^{2n+1} \cos \frac{2\pi}{3} + \left(-\frac{1}{2} \right)^n$$

AG

A1

3

[14]

Q8.

(a) (i) $\left(e^{\frac{2\pi i}{7}}\right)^7 = e^{2\pi i} = 1$

Or $z^7 = e^{2k\pi i} \quad z = e^{\frac{2k\pi i}{7}} \quad k = 1$

B1

1

(ii) Roots are $\omega^2, \omega^3, \omega^4, \omega^5, \omega^6$

OE; M1A0 for incomplete set

SC B1 for a set of correct roots in terms of $e^{i\theta}$

M1A1

2

(b) Sum of roots considered

M1

= 0

or $\sum_{r=0}^6 \omega^r = \frac{\omega^7 - 1}{\omega - 1} = 0$

A1

2

(c) (i) $\omega^2 + \omega^5 = e^{\frac{4\pi i}{7}} + e^{\frac{10\pi i}{7}}$

M1

$= e^{\frac{4\pi i}{7}} + e^{\frac{-4\pi i}{7}}$

Or $\cos \frac{4\pi}{7} + i \sin \frac{4\pi}{7} + \cos \frac{4\pi}{7} - i \sin \frac{4\pi}{7}$

A1

$= 2 \cos \frac{4\pi}{7}$

AG

A1

3

(ii) $\omega + \omega^6 = 2\cos\frac{2\pi}{7}; \quad \omega^3 + \omega^4 = 2\cos\frac{6\pi}{7}$

Allow these marks if seen earlier in the solution

B1,B1

Using part (b)

M1

Result

AG

A1

4

[12]

Q9.

(a) $(\cos \theta + i \sin \theta)^{k+1} = (\cos k\theta + i \sin k\theta)(\cos \theta + i \sin \theta)$

M1

Multiply out

Any form

A1

$= \cos(k+1)\theta + i \sin(k+1)\theta$

Clearly shown

A1

True for $n = 1$ shown

B1

$P(k) \Rightarrow P(k+1)$ and $P(1)$ true

provided previous 4 marks earned

E1

5

(b) $\frac{1}{z^n} = \frac{1}{\cos n\theta + i \sin n\theta} = \cos n\theta - i \sin n\theta$

or $z^{-n} = \cos(-n\theta) + i \sin(-n\theta)$

$(\cos \theta + i \sin \theta)^{-n}$

SC *quoted as* $\cos n\theta - i \sin n\theta$

earns M1A1 only

M1A1

$$z^n + \frac{1}{z^n} = 2 \cos n\theta$$

AG

A1

3

(c) $z + \frac{1}{z} = \sqrt{2}$
 $2 \cos \theta = \sqrt{2}$

M1

$$\theta = \frac{\pi}{4}$$

A1

$$z^{10} + \frac{1}{z^{10}} = 2 \cos\left(\frac{10\pi}{4}\right)$$

M0 for merely writing

$$z^{10} + \frac{1}{z^{10}} = 2 \cos 10\theta$$

M1

$$= 0$$

A1F

4

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[12]

Q10.

(a) (i) $\cos 3\theta + i \sin 3\theta = (\cos \theta + i \sin \theta)^3$

M1

$$= \cos^3 \theta + 3i \cos^2 \theta \sin \theta + 3i^2 \cos \theta \sin^2 \theta + i^3 \sin^3 \theta$$

A1

Real parts: $\cos 3\theta = \cos^3 \theta - 3 \cos \theta \sin^2 \theta$

AG

A1

3

(ii) Imaginary parts:
 $\sin 3\theta = 3 \cos^2 \theta \sin \theta - \sin^3 \theta$

A1F

1

(iii) $\tan 3\theta = \frac{\sin 3\theta}{\cos 3\theta}$
Used

M1

$$= \frac{3\cos^2 \theta \sin \theta - \sin^3 \theta}{\cos^3 \theta - 3\sin^2 \theta \cos \theta}$$

Error in sin 3θ

A1F

$$= \frac{3\tan \theta - \tan^3 \theta}{1 - 3\tan^2 \theta}$$

$$= \frac{\tan^3 \theta - 3\tan \theta}{3\tan^2 \theta - 1}$$

AG

A1

3

(b) (i) $\tan \frac{3\pi}{12} = 1$
Used (possibly implied)

B1

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$\tan \frac{\pi}{2}$ is a root of $1 = \frac{x^3 - 3x}{3x^2 - 1}$

Must be hence

M1

$$x^3 - 3x^2 - 3x + 1 = 0$$

A1

3

(ii) Other roots are $\tan \frac{5\pi}{12}$, $\tan \frac{9\pi}{12}$

B1B1

2

(c) $\tan \frac{\pi}{12} + \tan \frac{5\pi}{12} + \tan \frac{9\pi}{12} = 3$
Must be hence

M1

$$\tan \frac{\pi}{12} + \tan \frac{5\pi}{12} = 4$$

A1

2

[14]

Q11.

(a) (i) $\left(z + \frac{1}{z}\right)\left(z - \frac{1}{z}\right) = z^2 - \frac{1}{z^2}$

B1

1

(ii) $\left(z^2 - \frac{1}{z^2}\right)^2 \left(z + \frac{1}{z}\right)^2$
 $= \left(z^4 - 2 + \frac{1}{z^4}\right)\left(z^2 + 2 + \frac{1}{z^2}\right)$

Alternatives for M1A1:

$$\left(z^4 + 4z^2 + 6 + \frac{4}{z^2} + \frac{1}{z^4}\right)\left(z^2 - 2 + \frac{1}{z^2}\right) \text{ or}$$

$$\left(z^3 - \frac{1}{z^3}\right)^2 - 2\left(z^3 - \frac{1}{z^3}\right)\left(z - \frac{1}{z}\right) + \left(z - \frac{1}{z}\right)^2$$

M1A1

$$= z^6 + \frac{1}{z^6} + 2\left(z^4 + \frac{1}{z^4}\right) - \left(z^2 + \frac{1}{z^2}\right) - 4$$

CAO (not necessarily in this form)

A1

3

(b) (i) $z^n + \frac{1}{z^n} = \cos n\theta + i \sin n\theta + \cos(-n\theta) + i \sin(-n\theta)$

M1A1

$$= 2 \cos n\theta$$

AG

SC: if solution is incomplete and

$(\cos \theta + i \sin \theta)^{-n}$ is written as

$\cos n\theta - i \sin n\theta$, award M1A0A1

A1

3

(ii) $z^n - z^{-n} = 2i \sin n\theta$

B1

1

- (c) $\text{RHS} = 2 \cos 6\theta + 4 \cos 4\theta - 2 \cos 2\theta - 4$
ft incorrect values in (a)(ii) provided they are cosines

M1A1F

$$\text{LHS} = -64 \cos^4 \theta \sin^2 \theta$$

M1

$$\cos^4 \theta \sin^2 \theta$$

$$= -\frac{1}{32} \cos 6\theta - \frac{1}{16} \cos 4\theta + \frac{1}{32} \cos 2\theta + \frac{1}{16}$$

A1

4

(d) $-\frac{\sin 6\theta}{192} - \frac{\sin 4\theta}{64} + \frac{\sin 2\theta}{64} + \frac{\theta}{16} (+k)$

ft incorrect coefficients but not letters A, B, C, D

M1A1F

2

[14]

Q12.

- (a) Assume true for $n = k$
 $(\cos \theta + i \sin \theta)^{k+1}$
 $= (\cos k\theta + i \sin k\theta)(\cos \theta + i \sin \theta)$

M1

Multiply out

Any form

A1

$$= \cos(k+1)\theta + i \sin(k+1)\theta$$

A1

True for $n = 1$ shown

B1

$$P(k) \Rightarrow P(k+1) \text{ and } P(1) \text{ true}$$

Allow E1 only if previous 4 marks earned

E1

5

(b) $\left(\cos \frac{\pi}{6} + i \sin \frac{\pi}{6}\right)^6 = \cos \frac{6\pi}{6} + i \sin \frac{6\pi}{6}$

M1

$= -1$

A1

2

(c) $(\cos \theta + i \sin \theta)(1 + \cos \theta - i \sin \theta)$

M1

$= \cos \theta + \cos^2 \theta - i \sin \theta \cos \theta + i \sin \theta + i \sin \theta \cos \theta + \sin^2 \theta$

(Accept $-i^2 \sin^2 \theta$)

Or $e^{i\theta}(1 + e^{-i\theta})$

A1

$= 1 + \cos \theta + i \sin \theta$

AG

A1

3

(d) $\theta = \frac{\pi}{6}$ used

In the context of part (c)

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M1

Part (c) raised to power 6

M1

Use of result in part (b)

A1

$\left(1 + \cos \frac{\pi}{6} + i \sin \frac{\pi}{6}\right)^6 + \left(1 + \cos \frac{\pi}{6} - i \sin \frac{\pi}{6}\right)^6 = 0$

AG

A1

4

[14]

Q13.

$(\cos \theta + i \sin \theta)^{15} = \cos 15 \theta + i \sin 15 \theta$

or $= e^{15i\theta}$

M1

$$\cos 15\theta = 0$$

$$\sin 15\theta = -1$$

$$\text{or } -j = e^{\frac{3\pi i}{2}}$$

m1A1

$$15\theta = \frac{3\pi}{2} \text{ or } 270^\circ$$

m1 for **both** R&I parts written down

A1F

$$\theta = \frac{\pi}{10} \text{ or } 18^\circ$$

ft provided the value of 15θ is a correct value

A1F

5

SC

$$\cos 15\theta + i \sin 15\theta = i$$

(M1)

$$\sin 15\theta = -1$$

or for $\cos 15\theta = 0$

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(B1)

$$\theta = \frac{\pi}{10}$$

(B1)

(3)

[5]

Q14.

$$(a) \quad (i) \quad z + \frac{1}{z} = \cos \theta + i \sin \theta + \cos (-\theta) + i \sin (-\theta)$$

$$\text{Or } z + \frac{1}{z} = e^{i\theta} + e^{-i\theta}$$

M1

$$= 2\cos \theta$$

AG

A1

2

$$(ii) \quad z^2 + \frac{1}{z^2} = \cos 2\theta + i \sin 2\theta + \cos(-2\theta) + i \sin(-2\theta)$$

M1

$$= 2 \cos 2\theta$$

OE

A1

2

$$(iii) \quad z^2 - z + 2 - \frac{1}{z} + \frac{1}{z^2}$$

$$= 2 \cos 2\theta - 2 \cos \theta + 2$$

M1

$$\text{Use of } \cos 2\theta = 2 \cos^2 \theta - 1$$

m1

$$= 4 \cos^2 \theta - 2 \cos \theta$$

AG

A1

3

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$$(b) \quad z + \frac{1}{z} = 0 \quad z = \pm i$$

M1A1

$$z + \frac{1}{z} = 1 \quad z^2 - z + 1 = 0$$

M1A1

$$z = \frac{1 \pm i\sqrt{3}}{2}$$

A1F

Alternative:

$$\cos \theta = 0 \quad \theta = \pm \frac{1}{2} \pi$$

M1

$$z = \pm i$$

A1

$$\cos \theta = \frac{1}{2} \quad \theta = \pm \frac{1}{3} \pi$$

M1

$$z = e^{\pm \frac{1}{3}\pi i} = \frac{1}{2}(1 \pm i\sqrt{3})$$

A1A1

5

Accept solution to (b) if done otherwise

Alternative

$$\text{If } \theta = +\frac{1}{2}\pi \quad \theta = \frac{1}{3}\pi$$

M1

$$z = i \quad z = \frac{1 + \sqrt{3}i}{2}$$

A1

Or any correct z values of θ

M1

Any 2 correct answers

A1

One correct answer only

B1

EXAM PAPERS PRACTICE [12]

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Examiner reports

Q2.

- (a) (i) It had been expected that students would have written a correct identity using de Moivre's theorem at the beginning of their solution, but only the best students wrote $(\cos \theta + i \sin \theta)^4 = \cos 4\theta + i \sin 4\theta$ as a starting point. The binomial expansion was usually correct, although many did not use brackets correctly to indicate powers of i . The printed answer for $\cos 4\theta$ helped many to recover from sign errors in their expansion, though at least one " i " was often left in the expression offered for $\sin 4\theta$.
- (ii) Most students wrote down an expression for $\tan 4\theta$ by writing the quotient of their previous results for $\sin 4\theta$ and $\cos 4\theta$. Many students failed to justify the leap from this expression to the printed expression involving $\tan \theta$. It was necessary to indicate that the numerator and denominator had been divided throughout by $\cos^4 \theta$. Nevertheless, the majority of students scored full marks for both sections of part (a) and the printed answers clearly helped.
- (b) The crux of the proof was in realizing that $\tan 4\theta = 1$ when $\theta = \frac{\pi}{16}$ so as to obtain the given quartic equation in t . Very few were able to give a convincing reason as to why $\tan \frac{\pi}{16}$ was a root of the equation in t . No doubt influenced by the printed solution in the final part of the question, most gave their values of the other roots as $\tan \frac{3\pi}{16}$, $\tan \frac{5\pi}{16}$ and $\tan \frac{7\pi}{16}$ instead of the correct expressions involving trigonometric functions.
- (c) It was necessary for students to state explicitly that $\tan \frac{9\pi}{16} = -\tan \frac{7\pi}{16}$ and that $\tan \frac{13\pi}{16} = -\tan \frac{3\pi}{16}$ in order to score full marks. It was also necessary for examiners to see the sum of the roots written as -4 and the sum of the products of pairs of roots equal to -6 whilst showing that the sum of the squares of the roots was equal to 28 . Very few students scored full marks in this part of the question which was targeted at the very best students. Either the question had proved very difficult or students had run out of time at this stage of the examination.

Q3.

Although this question could be answered by a variety of methods, most candidates chose to clear fractions and equate real and imaginary parts, thus being able to score some marks for the question. However, many solutions either petered out at this point or continued with incorrect or unjustified assumptions, not realising that, for instance, $\cos(p\pi/8)$ was equal to $\sin(\pi/2 - p\pi/8)$; a simple step which, when spotted, almost invariably led to a completely correct solution. Whichever method a candidate used it was not possible to obtain complete credit without the introduction of $\pi/2$ at some stage.

Q4.

Responses to this question were extremely poor and very few candidates were able to make any headway at all after part (a). Very few candidates seemed to understand what

linear factors were – most merely wrote down the roots of z^5 in some other form. This in turn meant that virtually no one was able to answer part (c). There were not many serious attempts at part (d) in spite of the fact that it could be done by assuming the result in part (c), although some candidates were able to pick up the odd mark.

Q5.

Overall, this question was answered quite well. Part (a) was usually correct. In part (b)(i) a number of students gave their answer as $16 \cos^4 2\theta$ instead of an equation in z although some were able to recover this mark in subsequent working. The common error in part

(b)(ii) was to replace $z^8 + \frac{1}{z^8}$ by $\cos 8\theta$ rather than $2 \cos 8\theta$ and if this occurred it did mean that in part (c), students would arrive at a trigonometric equation they were unable to solve, and in part (d) an integral for which they could not score full marks. One unusual feature of part (d) was that a significant number of students, when attempting the integral, used the RHS of the equation in part (c) as their integrand.

Q6.

Generally candidates were well drilled in the book work of this question and consequently part (a) was a good source of marks for all, although in part (a)(ii) some solutions lacked intermediate steps between an expression for $\tan 5\theta = \sin 5\theta / \cos 5\theta$ and the printed answer. Responses to part (b) were not always very convincing although most realised it was an application of the result of part (a)(ii). The roots in part (b) were nearly always given as $\tan(r\pi/5)$ $r = 1, 2, 3, 4$, and this generally hindered progress in part (c) with candidates not realising that $\tan(4\pi/5)$, for instance, was equal to $-\tan(\pi/5)$. Only a handful of candidates rejected the $-\sqrt{5}$ when taking the square root in the final part.

Q7.

This question provided a good source of marks for many candidates. The commonest error in part (a) was to write $\alpha\beta\gamma$ as -16 , overlooking the fact that the coefficient of x^3 was not unity and, of course, leading to an incorrect value for γ . This error perpetuated in part (a)(ii) with $\alpha + \beta + \gamma$ written as p or $-p$ and the same for q .

Parts (b)(i) and (b)(ii) were generally well done, although it should be stated that when answers are printed, candidates are expected to provide sufficient detail to show clearly how their answers are arrived at. Part (b)(iii) was also quite well done and it was pleasing

to note that in some cases where candidates had not arrived at $\gamma = -\frac{1}{2}$, they went back to part (a)(ii) to identify their error. Just occasionally in part (b)(iii), some candidates made

blatant errors in their attempt to convert $4^x \cos \frac{p\pi}{3} + 4^x \cos \frac{p\pi}{3}$ into $2^{2x+1} \cos \frac{p\pi}{3}$.

Q8.

Whilst part (a) was generally well done, relatively few candidates expressed the other roots in terms of ω , but rather gave them in the form $re^{i\theta}$.

Few, also, were able to complete part (b). The relation $1 + \omega + \omega^2 = 0$ appeared with regularity.

Part (c) was poorly answered. Although correct answers were written down as they were

given in the question, few responses were convincing and as has already been stated earlier, if answers are given, it is the responsibility of the candidates to supply sufficient working to convince the examiner that they understand the methods involved.

Q9.

The general rules applicable to proof by induction in part (a) were usually understood, but because candidates realised that the product of $\cos\theta + i\sin\theta$ with $\cos k\theta + i\sin k\theta$ had to result in $\cos(k+1)\theta + i\sin(k+1)\theta$, many lost marks through omitting some of the intermediate steps.

In part (b), many candidates lost a mark by assuming that $(\cos\theta + i\sin\theta)^{-n}$ was equal to $\cos n\theta - i\sin n\theta$ without any justification.

Part (c) was almost invariably correctly done.

Q10.

Responses to this question were rather disappointing. In part (a)(i), although most candidates correctly quoted $\cos 3\theta + i\sin 3\theta = (\cos\theta + i\sin\theta)^3$, some immediately went on to use the multiple angle formulae instead of expanding $(\cos\theta + i\sin\theta)^3$. Some of those candidates who expanded $(\cos\theta + i\sin\theta)^3$ did not seem to realise that the answers to parts (a)(i) and (a)(ii) were obtained by simply equating real and imaginary parts.

Other candidates wrote i^3 as $+i$ and so were unable to reach the correct result of part (a)(ii) and the printed result in part (a)(iii). Even those candidates who worked parts (a)(i) and

(a)(ii) correctly in terms of $\sin\theta$ and $\cos\theta$, having written $\frac{\sin 3\theta}{\cos 3\theta}$, did not realise that the division of numerator and denominator by $-\cos^3\theta$ would give the printed result, but rather chose to use $\sin^2\theta + \cos^2\theta = 1$ to express numerator and denominator in a different form, with no hope of reaching the printed result. Although the question in part (b)(i) started with

the word 'hence' few candidates took up the hint and replaced θ by $\frac{\pi}{12}$ in part (a)(iii).

If this part was attempted it was often done by solving the cubic equation in x to find its

three roots and then by quoting that $\tan \frac{\pi}{12} = 2 - \sqrt{3}$ and a corresponding result for $\tan \frac{5\pi}{12}$ and consequently using these results in part (c), a method not indicated by the question.

Q11.

The algebraic manipulation in part (a)(ii) of this question was poor, especially from those candidates who failed to use the hint of part (a)(i) – a hint intended to simplify the algebra. Sign errors were frequent and in many cases led to multiples of sine rather than cosine in part (c) in spite of the form given in part (c). Many candidates did not score full marks in part (b)(i) due to their quoting that $(\cos\theta + i\sin\theta)^{-n}$ was $\cos n\theta - i\sin n\theta$, thus omitting the important fact that $(\cos\theta + i\sin\theta)^{-n} = \cos(-n\theta) + i\sin(-n\theta)$ from de Moivre's theorem. Part (c) was rarely fully correct. Common mistakes were either to assume that $\cos^4\theta \sin^2\theta$ was equal to the printed result with numbers substituted for A , B , C and D

thus ignoring $\left(z + \frac{1}{z}\right)^4 \left(z - \frac{1}{z}\right)^2$ completely, or by thinking that for instance $z^6 + \frac{1}{z^6}$ was equal to $\cos 6\theta$ rather than $2\cos 6\theta$, leaving out the factor 2 altogether.

Q12.

It was clear that a good number of candidates had not met the proof of de Moivre's Theorem by induction before and there were not many solutions gaining full marks. It was also not uncommon to see expressions such as

$$\cos k\theta + i \sin k\theta + \cos \theta + i \sin \theta = \cos (k+1)\theta + i \sin (k+1)\theta.$$

Parts (b) and (c) were generally well done, although in part (c) a number of candidates, when multiplying $i \sin \theta$ by $-i \sin \theta$, wrote $-i \sin^2 \theta$ and thus were unable to complete this part satisfactorily. Those candidates who spotted the connection between parts (c) and (d) usually went on to write out a correct solution to part (d), but it was disappointing to see

$$\left(1 + \cos \frac{\pi}{6} + i \sin \frac{\pi}{6}\right)^6 \text{ written as } i^6 + \cos \frac{6\pi}{6} + i \sin \frac{6\pi}{6} \text{ with alarming regularity.}$$

Q13.

There were many incomplete solutions to this question. Whilst most candidates used the de Moivre's Theorem correctly, many candidates either equated real parts only to arrive at an incorrect answer, or equated imaginary parts. In this latter case, the solution

$$\theta = -\frac{\pi}{30} \text{ appeared frequently in spite of the request in the question that } \theta \text{ should be positive, or the correct answer appeared but from an incomplete solution. Some candidates solved } \cos \theta = 0 \text{ and } \sin \theta = -1 \text{ but gave two different values of } \theta \text{ as their answer, one from each equation.}$$

Q14.

Parts (a) was quite well done and many candidates scored the available seven marks. There were however some serious algebraic errors, the commonest of which was

$$\text{to equate } z^2 + \frac{1}{z^2} \text{ to } \left(z + \frac{1}{z}\right)^2 \text{ in part (a)(ii) with some consequent faking to arrive at the printed result in part (a)(iii). Part (b) however was very poorly attempted. Candidates did not seem to realise that } z \text{ could be equal to zero and consequently multiplied } (\cos \theta + i \sin \theta)^2 \text{ by } 4\cos^2 \theta - 2\cos \theta. \text{ Of those candidates who realised that } 4\cos^2 \theta - 2\cos \theta \text{ was equal to zero, the factorisation of this quadratic in } \cos \theta \text{ evaded most and, even when attempts were made to solve } 4\cos^2 \theta - 2\cos \theta = 0, \text{ the factor } \cos \theta \text{ disappeared and the}$$

other solution $\cos \theta = 1/2$ usually produced just one root from $\theta = \frac{\pi}{3}$. Candidates who were able to obtain both $\cos \theta = 0$ and $\cos \theta = 1/2$ usually produced only two solutions and subsequently two roots. It did not seem to occur to candidates that a quartic equation would have four roots.